Gamma Distribution

The gamma distribution is frequently the prob. model for waiting times, for instances, in life testing, the waiting time until "death" is the random variable which frequently has a gamma distribution.

Since the two parameters α and β provide a great deal of flexibility.

The p.d.f of the Gamma distribution is

$$f(x) = \frac{1}{\Gamma\alpha\, \beta^{\alpha}} x^{\alpha-1} e^{-\frac{x}{\beta}} \qquad 0 < x < \infty$$

$$\alpha > 0,\ \beta > 0$$

$$\text{or } f(x) = \frac{1}{\Gamma\alpha+1\, \beta^{\alpha+1}} x^{\alpha} e^{-\frac{x}{\beta}} \qquad \Gamma\alpha > 0$$

$$\Gamma\alpha+1 > 0$$

Question

Find the m.g.f. and cumulant generating function also find γ_1, γ_2, β_1 and β_2.

Sol: By definition

$$M_x(t) = E(e^{tx})$$

$$= \int_0^{\infty} e^{tx} \cdot \frac{1}{\Gamma\alpha\, \beta^{\alpha}} x^{\alpha-1} e^{-\frac{x}{\beta}}\, dx$$

$$= \frac{1}{\Gamma\alpha\, \beta^{\alpha}} \int_0^{\infty} x^{\alpha-1} e^{-x\left(\frac{1}{\beta}-t\right)}\, dx$$

put $x\left(\frac{1}{\beta}-t\right) = y$

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$$x\left(\frac{1-\beta t}{\beta}\right) = y \qquad x=0 \quad y=0$$

$$x = \left(\frac{\beta}{1-\beta t}\right) y \qquad x=\infty \quad y=\infty$$

$$dx = \frac{\beta}{1-\beta t}\, dy$$

$$M_x(t) = \frac{1}{\Gamma\alpha\, \beta^\alpha}\int_0^\infty \left(\frac{\beta y}{1-\beta t}\right)^{\alpha-1} e^{-y} \cdot \frac{\beta}{1-\beta t}\, dy$$

$$M_x(t) = \frac{\left(\frac{\beta}{1-\beta t}\right)^{\alpha-1} \frac{\beta}{(1-\beta t)}}{\Gamma\alpha\ \beta^\alpha}\int_0^\infty y^{\alpha-1} e^{-y}\, dy$$

$$= \frac{\left(\frac{\beta}{1-\beta t}\right)^{\alpha} \cancel{\Gamma\alpha}}{\cancel{\Gamma\alpha}\ \beta^\alpha}$$

$$= \frac{\cancel{\beta^\alpha}}{(1-\beta t)^\alpha\, \cancel{\beta^\alpha}} = \frac{1}{(1-\beta t)^\alpha}$$

$$\underline{M_x(t) = (1-\beta t)^{-\alpha}}$$ (Find Mean and Variance using M.g.f)

For cumulant g.f.

$$\ln M_x(t) = -\alpha \ln(1-\beta t)$$

$$= \alpha\left(-\ln(1-\beta t)\right)$$

Since $-\ln(1-x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots$

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$$\ln M_x(t) = \alpha \left\{ (\beta t) + \frac{(\beta t)^2}{2} + \frac{(\beta t)^3}{3} + \frac{(\beta t)^4}{4} + \cdots \right\}$$

By comparing the coefficients of $\frac{t^r}{r!}$

$$\mu_1' = K_1 = \alpha\beta$$

$$K_2 = \alpha\beta^2$$

$$K_3 = 2\alpha\beta^3$$

$$K_4 = 6\alpha\beta^4$$

$$\gamma_1 = \frac{K_3}{K_2^{3/2}} \Rightarrow \frac{2\alpha\beta^3}{(\alpha\beta^2)^{3/2}}$$

$$= \frac{2\alpha\cancel{\beta^3}}{\alpha^{3/2}\cancel{\beta^3}}$$

$$= \frac{2}{\sqrt{\alpha}}$$

$$\gamma_2 = \frac{K_4}{K_2^2} \Rightarrow \frac{6\alpha\beta^4}{(\alpha\beta^2)^2}$$

$$\gamma_2 = \frac{6}{\alpha}$$

Moments about mean

$$\mu_1 = 0$$

$$\mu_2 = K_2 \Rightarrow \alpha\beta^2$$

$$\mu_3 = K_3 = 2\alpha\beta^3$$

$$\mu_4 = K_4 + 3K_2^2$$

$$\Rightarrow 6\alpha\beta^4 + 3(\alpha\beta^2)^2$$

$$\Rightarrow 6\alpha\beta^4 + 3\alpha^2\beta^4$$

$$\mu_4 \Rightarrow \alpha\beta^4(6 + 3\alpha)$$

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$$\beta_1 = \frac{(\mu_3)^2}{(\mu_2)^3} \Rightarrow \frac{(2\alpha\beta^3)^2}{(\alpha\beta^2)^3} \Rightarrow \frac{4\alpha^2\beta^6}{\alpha^3\beta^6}$$

$$\beta_1 = \frac{4}{\alpha} \qquad \sqrt{\beta_1} = \gamma_1$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} \Rightarrow \frac{\alpha\beta^4(6+3\alpha)}{(\alpha\beta^2)^2}$$

$$\Rightarrow \frac{\alpha\beta^4(6+3\alpha)}{\alpha^2\beta^4} \Rightarrow \frac{6\alpha+3\alpha^2}{\alpha^2}$$

$$\beta_2 = \frac{6}{\alpha} + 3$$

$$\beta_2 = \gamma + 3$$

Question

Find the characteristic function of the Gamma Distribution, Also find the first four centeral Moments.

Sol

$$\phi_x(t) = (1-\beta i t)^{-\alpha}$$

taking ln both sides

$$\ln \phi_x(t) = \alpha(-\ln(1-\beta i t))$$

By comparing the coefficients of $\frac{(it)^r}{r!}$ we get

K_1, K_2, K_3, K_4

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Question

Find the Mode of the gamma Distribution

Sol. The p.d.f. of the Gamma Distribution

$$f(x) = \frac{1}{\Gamma\alpha\, \beta^{\alpha}} x^{\alpha-1} e^{-x/\beta} \qquad 0 < x < \infty$$

Taking ln both sides.

$$\ln f(x) = \ln\left(\frac{1}{\Gamma\alpha\,\beta^{\alpha}}\right) - \frac{x}{\beta} + (\alpha-1)\ln x$$

Diff. w. r. to x

$$\frac{d}{dx} \ln f(x) = 0 - \frac{1}{\beta} + \frac{\alpha-1}{x}$$

equating it zero

$$-\frac{1}{\beta} + \frac{\alpha-1}{x} = 0$$

$$\frac{\alpha-1}{x} = \frac{1}{\beta}$$

$$\frac{x}{\alpha-1} = \beta$$

$$x = (\alpha-1)\beta$$

again diff. w. r. to x

$$\frac{d^2}{dx^2} \ln f(x) = -\frac{(\alpha-1)}{x^2} < 0$$

therefore Mode $= (\alpha-1)\beta$

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Show that incomplete gamma Distribution

$$f(y) = \frac{1}{\Gamma\alpha} e^{-y} y^{\alpha-1} \qquad 0 < y < \infty.$$

Sol.: We know that

$$f(x) = \frac{1}{\Gamma\alpha\, \beta^\alpha} x^{\alpha-1} e^{-\frac{x}{\beta}} dx \qquad 0 < x < \infty$$

put $\frac{x}{\beta} = y$ $\qquad$ $x=0$, $y=0$ | $x=\infty$, $y=\infty$.

$x = \beta y$

$dx = \beta dy$

Now.

$$f(y) = \frac{1}{\Gamma\alpha\, \beta^\alpha} (\beta y)^{\alpha-1} e^{-y} . \beta dy$$

$$= \frac{1}{\Gamma\alpha\, \cancel{\beta^\alpha}} y^{\alpha-1} \cancel{\beta^\alpha} e^{-y} dy$$

$$f(y) = \frac{1}{\Gamma\alpha} e^{-y} y^{\alpha-1} \qquad 0 < y < \infty$$

Question Show that for the Distribution

$$f(x) = \frac{1}{\Gamma m} e^{-x} x^{m-1} \qquad 0 < x < \infty,\ m > 0$$

Mean and Variance are equal. also show that $\mu'_r = m(m+1)(m+2) - - - - (m+r-1)$

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